Django LMS - Project Report

Learning Management System with Quizzes and Progress Tracking

1. Abstract

This report describes the design and development of a web-based Learning Management System (LMS) built with Django. The system allows teachers to publish courses made of modules and lessons, and allows students to enroll, learn, track their progress, and take quizzes that are graded automatically. Access is controlled by user roles, so each user only sees the features that belong to them.

2. Objectives

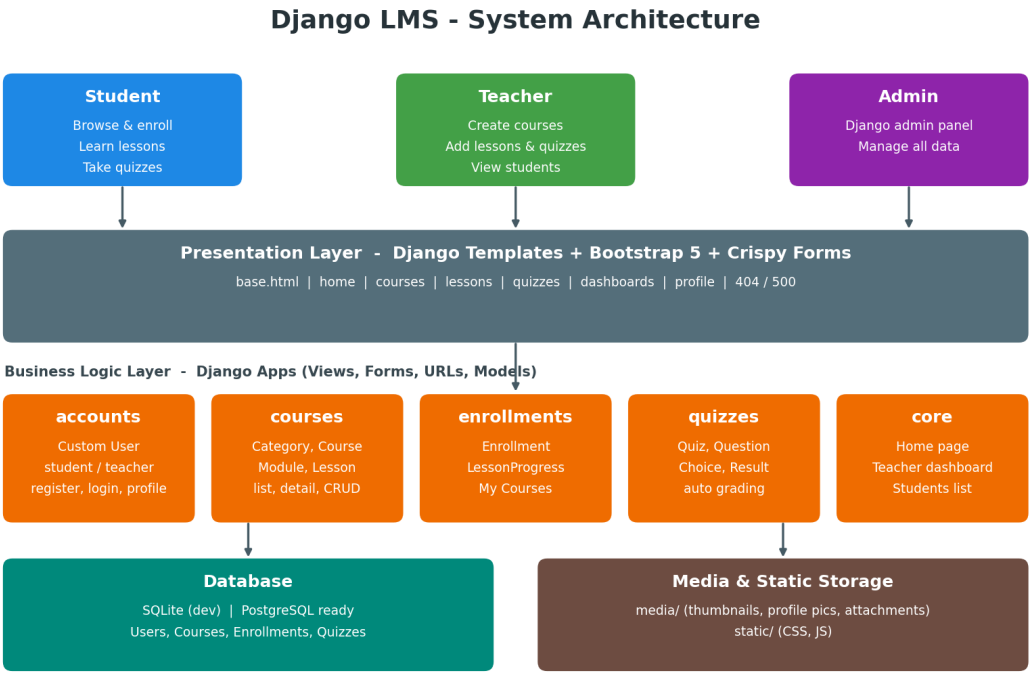
- Provide a simple platform where teachers can create and manage online courses.
- Let students enroll in courses and learn at their own pace.
- Track each student's progress lesson by lesson.
- Assess learning through multiple-choice quizzes with instant results.
- Keep the system secure with role-based access control.

3. Technology Stack

Layer	Technology
Language	Python 3.11
Framework	Django 5.2
Frontend	Django Templates, Bootstrap 5
Forms	django-crispy-forms, crispy-bootstrap5
Database	SQLite (PostgreSQL ready)
Image handling	Pillow
Configuration	python-decouple (.env file)

4. System Architecture

The project follows Django's MVT (Model, View, Template) pattern and is divided into five apps, each with a single responsibility.

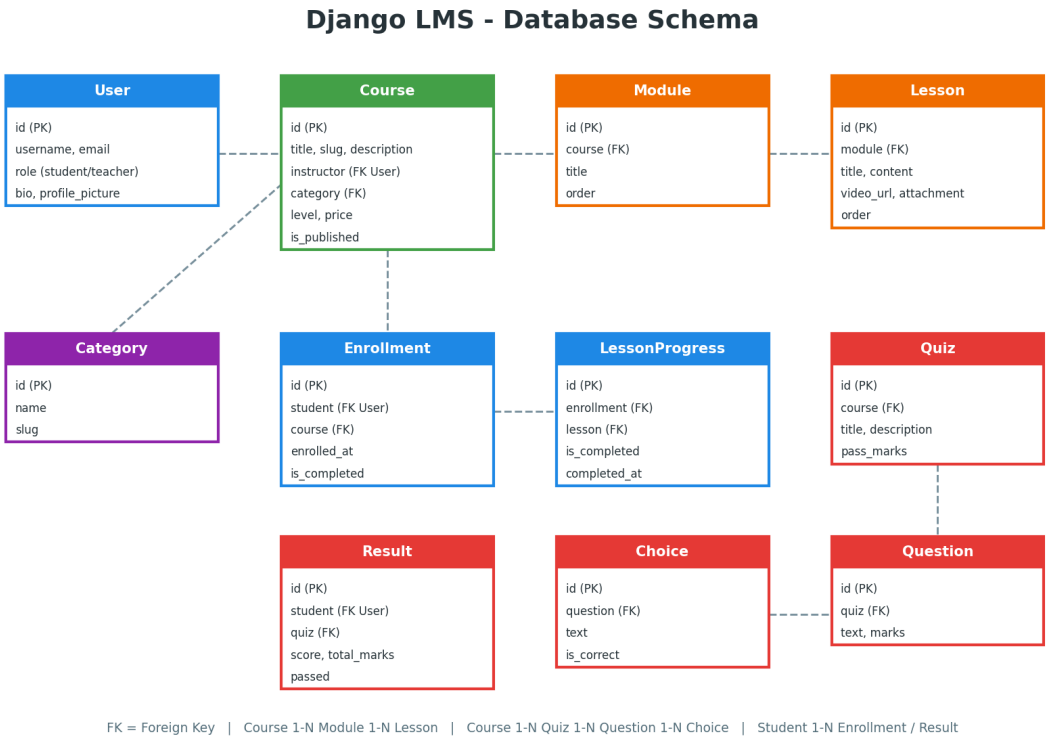


System architecture

App	Responsibility
accounts	Custom user model, registration, login, logout, profile
courses	Categories, courses, modules, lessons, teacher course management
enrollments	Enrollment, lesson progress, My Courses page
quizzes	Quizzes, questions, choices, results, automatic grading
core	Home page, teacher dashboard, student list per course

5. Database Design

The database contains 11 tables. A custom `User` model extends Django's `AbstractUser` with a `role` field (student or teacher), a bio, and a profile picture.



Database schema

Key relationships:

- A course belongs to one teacher and one category.
- A course has many modules, and a module has many lessons.
- An enrollment links one student to one course; a student can enroll in a course only once.
- Lesson progress records which lessons an enrollment has completed.
- A quiz belongs to a course and has many questions; each question has several choices, one of them correct.
- A result stores a student's score and pass/fail status for one quiz attempt.

6. Features

6.1 Student features

- Registration, login, logout, and profile editing
- Course search and category filter
- Enrollment and lesson viewing (text, video link, attachments)
- Mark lessons complete; progress percentage shown on My Courses
- Take quizzes and see score with pass/fail result
- My Results page with all attempts

6.2 Teacher features

- Dashboard listing courses, publish status, and enrolled student count
- Create, edit, and delete courses
- Add modules and lessons
- Create quizzes and add multiple-choice questions
- View enrolled students with their progress

6.3 Administration

All models are registered in the Django admin panel, with filters, search, and inline editing for modules, lessons, questions, and choices.

7. Security and Access Control

- Passwords are hashed by Django's authentication system.
- All forms use CSRF protection.
- Pages are protected with `login_required` and role checks.
- Only enrolled students (or the course teacher) can open lessons.
- Only enrolled students can take quizzes.
- Teachers can edit only their own courses, modules, lessons, and quizzes.
- Delete and logout actions accept POST requests only.
- The secret key and debug flag are stored in a `.env` file that is excluded from version control.

8. Testing

The system was tested manually using the following checklist:

Area	Checks
Public visitor	Home page, course list, course detail, login prompt for enrolling
Student	Register, enroll, complete lessons, progress bar, quiz, results, profile
Teacher	Create course, module, lesson, quiz with questions; students list; edit and delete
Access control	Student blocked from teacher pages; lessons blocked without enrollment
Admin	All models visible and editable
Error pages	Custom 404 and 500 pages

9. Limitations

- Quizzes can be attempted any number of times.
- Only multiple-choice questions with a single correct answer are supported.
- Courses are free to enroll in; there is no payment system.
- No email notifications.
- Automated tests are not yet written.

10. Future Work

- Quiz attempt limits and time limits
- Course reviews and ratings
- Email notifications
- Payment gateway integration
- REST API with Django REST Framework
- Automated unit tests
- Deployment on a cloud platform with PostgreSQL

11. Conclusion

The project delivers a working LMS covering the full learning cycle: course creation, enrollment, learning, progress tracking, and assessment. Its modular structure makes it easy to extend with new features such as payments, ratings, and APIs.